

GSOC 2026 MID-TERM PROGRESS

ADD SUPPORT FOR PREPARED STATEMENTS IN LUASQL (OCI8)

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PROGRESS OVERVIEW

Completed

- Common Prepared Statement API finalized
- OCI8 stmt_data
- LuaSQL Datatype System (`luasql.type.*`)
- `conn:prepare()`
- `stmt:close()`
- Test additions and fixes
- Code review improvements

Remaining

- `stmt:execute(params)`
 - Parameter validation
 - Parameter binding
 - DML execution
 - SELECT + cursor integration
- (If time permits)
 - PostgreSQL implementation

MAJOR DESIGN DECISIONS

1. Removed `stmt:bind()`

- Originally

```
stmt:bind(...)  
stmt:bind(...)  
stmt:execute()
```

- Now

```
stmt:execute({  
    empno = {101, luasql.type.int},  
    ename = {"Risho", luasql.type.string}  
})
```

- Example for Positional Binding

```
stmt:execute({  
    [1] = {101, luasql.type.int},  
    [2] = {"Risho", luasql.type.string}  
})
```

2. Common LuaSQL Type System

- Instead of exposing

```
SQLT_INT  
MYSQL_TYPE_LONG  
INT8OID
```

- Introduced

```
luasql.type.int  
luasql.type.number  
luasql.type.string  
...
```

3. Explicit Type Conversion

- Example

```
id   = {21,   luasql.type.int},  
salary = {95000.5, luasql.type.number},  
active = {true,  luasql.type.boolean},  
name  = {"Risho", luasql.type.string},  
joined = {"2026-05-29 19:30:00", luasql.type.timestamp},  
dob   = {"2026-05-29", luasql.type.date},  
time  = {"19:30:00", luasql.type.time},  
dept  = luasql.type.null
```

CONN:PREPARE()

- `stmt = conn:prepare(sql)`

Implementation

- OCISstmtPrepare2
- Statement type detection
- Dedicated OCI error handle
- Statement userdata
- Registry references
- Statement counter

Also improved

- Allocation order
- Cleanup on allocation failure
- Memory safety

STMT:CLOSE()

- Stmt:close()

Responsibilities

- Release OCI statement
- Free OCI error handle
- Update statement counters
- Prevent double free
- Prevent closing while cursor is active

Design discussions

- :close() vs __gc
- :close() vs __close
- Cursor ownership
- Forced cleanup

CODE REVIEW & IMPROVEMENTS

- Improved allocation order
- Fixed counter underflow
- Added helper functions
- Added `__index` metamethod for `luasql.type`
- Improved cleanup logic
- Added datatype tests
- Fixed OCI8 test failures

Testing

- Added datatype constant tests
- Fixed automatic cursor cleanup
- Investigated rollback issues
- Removed inconsistent `numrows()` implementation
- Added OCI8-specific test comments

THANK YOU

Any Questions?